12	**Answer: C**
	Solution $x = -x_o \cos \omega t$ Considering equilibrium at mid-tide $2x_o = 3.0 - 1.0 = 2.0\ \text{m}$ $x_o = 1.0\ \text{m}$ $x = -1.0 \cos \omega t$ $-0.5 = -1.0 \cos \omega t$ $\frac{2\pi}{12} t = \frac{\pi}{3}$ $t = 2.0\ \text{h}$ Max duration in harbour (x>-0.5) = 12 - 2(2.0) = 8 h